Disaster Recovery and Backup Runbook

Document ID	ENG-014
Department	Engineering
Document type	Runbook
Owner	Head of Site Reliability Engineering
Version	1.3
Status	APPROVED
Effective / last reviewed	2025-06-12
Audience	Engineering staff
Related documents	SEC-013, ENG-009, SEC-005, ENG-010

Summary. Runbook for backup, restore and disaster recovery of Northstar Systems production services, covering RPO and RTO by tier, backup schedules and retention, copies to the Hyderabad region, quarterly restore tests, twice-yearly failover drills and step-by-step failover and failback procedures.

SYNTHETIC DOCUMENT - Northstar Systems is a fictional company. This document exists for development, testing and demonstration only and is not real legal, employment, security or compliance advice.

1. Purpose and Scope

This runbook tells Site Reliability Engineering (SRE), service owners and on-call engineers how Northstar Systems Private Limited protects production data and restores production services after a failure. It defines recovery targets, backup and retention rules, the testing programme and the step-by-step procedures to follow when a service, a database or an entire region becomes unavailable.

The runbook applies to all production services hosted in the primary cloud region at Mumbai, with the secondary disaster recovery (DR) region at Hyderabad, including the Northstar Nexus Platform, Nexus Connect, Nexus Insight and the internal platforms that engineering operates. It does not cover the recovery of employee laptops, which is handled by IT, nor business continuity for offices, which is covered by the business continuity plan (see SEC-013). It should be used together with the incident process in ENG-004 and the monitoring standard in ENG-010.

2. Definitions

- **RPO (Recovery Point Objective)**: the maximum acceptable amount of data loss, measured as time. An RPO of 15 minutes means that at most the last 15 minutes of data may be lost.
- **RTO (Recovery Time Objective)**: the maximum acceptable time from declaring a disaster to restoring service.
- **MTD (Maximum Tolerable Downtime)**: the longest outage the business can accept for customer platform operations, set at 4 hours. RTO values must always be inside the MTD for services that customers depend on.
- **Full backup**: a complete copy of a database or volume.
- **Log shipping**: continuous copying of transaction logs so that a database can be restored to a point in time.

- **Failover**: moving production traffic from the Mumbai primary region to the Hyderabad DR region.
- **Failback**: returning traffic to Mumbai after it is healthy.
- **DR Commander**: the engineer who leads the recovery. For a regional disaster this is the Incident Commander of the SEV1 incident.
- **Service tiers**: Tier-1 (customer-facing, revenue critical), Tier-2 (internal critical) and Tier-3 (non-critical).

3. Roles and Responsibilities

Role	Responsibility
Head of SRE	Owns this runbook, approves drill plans and signs off drill results
Service owner	Keeps the service's recovery information up to date and fixes gaps found in tests
DBA (Data Engineering)	Runs database backup, restore and point-in-time recovery
DR Commander	Declares the disaster, directs failover and decides when to fail back
Communications Lead	Updates Northstar Status and informs Customer Success
Operations Lead	Executes the technical steps, with the Scribe recording actions
Security Desk	Joins if the disaster may be a security event; preserves evidence
COO	Leads the Crisis Management Team when the event affects the wider business

4. Recovery Objectives by Tier

Every service has a tier recorded in the service catalogue. The recovery objectives for each tier are fixed as follows.

Tier	RPO	RTO
Tier-1	15 minutes	1 hour
Tier-2	4 hours	8 hours
Tier-3	24 hours	72 hours

If a service's design cannot meet its tier's objectives, the service owner must either raise an exception through NorthDesk, approved by the Head of SRE for at most 90 days, or redesign the service. A service must not be described as Tier-1 in customer commitments unless its design can meet the Tier-1 objectives.

The Maximum Tolerable Downtime for customer platform operations is 4 hours. When a Tier-2 or Tier-3 service is needed to operate the customer platform, its practical recovery priority follows the customer impact, not the tier alone, and the DR Commander may reorder recovery.

5. Backup Standards

5.1 Schedules

Tier-1 databases receive a daily full backup and continuous log shipping at 15-minute intervals. The 15-minute log shipping is what makes the 15-minute RPO achievable. Tier-2 databases receive a daily full backup plus log shipping at intervals that meet the 4-hour RPO. Tier-3 data receives at least a daily backup.

Object storage and file-based data for Tier-1 services must be versioned and replicated to the secondary region. Configuration, infrastructure definitions and pipeline definitions must be held in version control and must not depend on backups of running hosts. Secrets are held in NorthVault or the KMS-backed secret store, and the process for restoring them is covered in section 9.

5.2 Retention

Backup type	Retention
Daily backups	35 days
Monthly backups	12 months
Log segments (Tier-1)	Long enough to cover point-in-time recovery across the 35-day daily window

The first backup of each calendar month is retained as the monthly backup. Backups that exceed their retention period must be deleted by the backup tool automatically. A legal hold applied by Legal overrides deletion. Backups containing personal data remain subject to erasure obligations under the privacy rules, and where a data subject erasure request has been approved by the Privacy Officer, the team must ensure that the data is not restored into production from older backups.

5.3 Copy to the secondary region and encryption

All backups are copied to the secondary region, Hyderabad. All backups must be encrypted, at rest with AES-256 using keys held in the cloud KMS, and in transit with TLS 1.2 or higher (TLS 1.3 preferred). The backup copy job must alert if a copy is more than 2 hours late for Tier-1 data. Backup encryption keys must not be stored in the same system as the backups, and KMS master key rotation every 12 months must not make older backups unreadable; the key service retains previous key versions for decryption.

Access to backups is restricted by role, uses just-in-time access under the access management rules, and every restore of Restricted data must be recorded in a NorthDesk ticket with the reason.

5.4 Backup monitoring

Backup success, duration, size and the age of the latest copy in Hyderabad are monitored in NorthWatch. A failed backup for a Tier-1 database must raise a ticket within 1 hour and must be fixed or escalated the same business day. Two consecutive failed daily backups for a Tier-1 database are escalated to the Head of SRE. A backup that completes but cannot be restored is treated as a failed backup, which is why restore tests are mandatory.

6. Testing Programme

6.1 Quarterly restore tests

A restore test must be run every quarter for each Tier-1 database, and should be run for Tier-2 databases at least twice a year. The test restores the most recent daily backup into an isolated environment, replays logs to a chosen point in time, runs data integrity checks (row counts, checksums on key tables, application smoke tests) and measures the time taken. The DBA records the result, the measured RTO and the measured data gap in a NorthDesk ticket. A restore that takes longer than the tier's RTO, or leaves a data gap larger than the RPO, is a failed test and must produce a corrective action with an owner and a due date within 30 days.

6.2 DR failover drills

A full DR failover drill is run twice a year, in April and October. In a drill, production Tier-1 traffic, or a representative portion of it agreed in advance, is moved to Hyderabad, the services run there for a stated period, and traffic is then failed back to Mumbai. Drill dates are announced to Customer Success at least 14 days in advance and are planned inside the standard production window of Tuesday to Thursday between 10:00 and 16:00 IST wherever the drill touches customer traffic, unless the Head of SRE and the CAB agree a different time. A drill must not take place during a freeze period.

For each drill the Head of SRE appoints a DR Commander and an observer from outside the SRE team. The drill report must record the elapsed time of each step, the achieved RTO and RPO, problems found and action items, and must be filed within 10 business days.

6.3 Tabletop exercises

Between drills, the SRE team runs at least one tabletop exercise per year in which the team walks through a scenario such as loss of the Mumbai region, accidental deletion of a production table or a ransomware event. Tabletops test communications and decision making and are not a substitute for technical drills.

7. When to Declare a Disaster

A disaster is declared when a Tier-1 service in the primary region cannot be restored to normal operation within a time that would still meet its RTO, or when the primary region is unavailable. In practice the on-call engineer opens a SEV1 incident in line with ENG-004 and the Incident Commander consults the Head of SRE. The decision to fail over to Hyderabad is made by the DR Commander using these guidelines.

- **Fail over** if the Mumbai region, or the networking or storage layer serving Tier-1 services, is down and the cloud provider's estimate for repair is longer than 30 minutes or is unknown, because a failover takes time to complete and the Tier-1 RTO is 1 hour.
- **Do not fail over** for a single-service fault that can be fixed faster by rollback or restart, or for a fault in application code that would be replicated to the DR region.
- **Restore instead of failover** for data corruption or accidental deletion, because replication would copy the damage. In this case use point-in-time recovery in section 8.3.

The DR Commander records the time of the declaration, because the RTO clock starts at the declaration.

8. Recovery Procedures

8.1 Regional failover to Hyderabad

1 The DR Commander declares the disaster, starts the incident bridge and assigns the Communications Lead, Operations Lead and Scribe.
2 The Communications Lead posts to Northstar Status within 30 minutes of the SEV1 declaration, and Customer Success sends the first customer notification to affected Enterprise and Premier contacts within the same 30 minutes. Status updates continue every 30 minutes for SEV1.
3 The Operations Lead confirms the latest replicated data in Hyderabad and notes the time of the last applied log. This time defines the actual data gap. The target is within 15 minutes for Tier-1 databases.

4 The DBA promotes the Hyderabad replica of each Tier-1 database to primary, in the order set in the service catalogue (identity and configuration stores first, then transactional databases, then analytics).
5 The Operations Lead scales the Hyderabad application tier to production size using the infrastructure definitions held in version control and deploys the last released version of each service. No new release should be deployed during a failover.
6 The Operations Lead validates internal health checks and runs the integration smoke tests for the Nexus Platform, including authentication, a test Pipeline run and a Webhook delivery.
7 The DR Commander approves the change of traffic. DNS and load balancer entries are updated to point to Hyderabad. Time-to-live values for the main entries are pre-set low to keep the switch fast.
8 The Operations Lead confirms from NorthWatch that error rate, latency and queue depth are within normal ranges. Tier-1 traffic must be restored within 1 hour of the declaration.
9 The Communications Lead announces restoration, and the Scribe records the measured RTO and any data gap.

8.2 Recovery of Tier-2 and Tier-3 services

After Tier-1 services are restored, the Operations Lead recovers Tier-2 services (target 8 hours from declaration) and then Tier-3 services (target 72 hours), starting with those that support the customer platform. Where Tier-2 and Tier-3 services had been backed up daily, the data gap may be up to 4 hours and 24 hours respectively, and the service owner must tell users of that gap. If reconstruction is impossible without going beyond a tier's recovery objective, the DR Commander informs the VP Engineering and the Head of the affected department.

8.3 Point-in-time recovery after corruption or deletion

1 Identify the time just before the damage occurred. The Scribe should note the source of that estimate, such as a deployment time or a log entry.
2 The DBA restores the last daily backup before that time into an isolated environment, then replays the shipped log segments up to the chosen point, a few minutes before the damage.
3 The service owner and the DBA verify the restored data with the integrity checks.
4 The DBA exports only the affected rows or tables and applies them to production, with two-person review, in line with the database change rules for data fixes.
5 The team records in NorthDesk what was restored, the data lost between the restore point and the incident and who approved it.

For Tier-1 databases, point-in-time recovery to any minute within the log retention window is possible because of the 15-minute log shipping and the daily full backup. If the damage is older than 35 days, only monthly backups are available, so the recovery granularity falls to one month and the DR Commander must inform the Privacy Officer if personal data is involved.

8.4 Failback to Mumbai

Failback is a planned change, not an emergency. It starts when the Mumbai region is confirmed stable for at least 24 hours. The steps are to re-establish replication from Hyderabad to Mumbai, wait for replicas to catch up, schedule failback inside the Tuesday to Thursday 10:00 to 16:00 IST window with a NorthDesk change ticket and a rollback plan, put the service into a short read-only period if needed, switch traffic and database primaries back to Mumbai, verify using the same checks as in section 8.1 and then restore the normal replication direction. The failback must be approved by the DR Commander and the Head of SRE.

9. Dependencies Needed for Recovery

Recovery fails if essential dependencies are unavailable, so these must be tested in each drill.

- **NorthID and NorthVault**: engineers need to authenticate and to retrieve secrets. A copy of the break-glass accounts must work without NorthID, monitored, with every use reviewed within 1 business day.
- **NorthGate**: production administration is only through the bastion with session recording. A bastion in Hyderabad must be available.
- **NorthShip**: the pipeline platform must be able to deploy the last released version to Hyderabad. Artefacts must be replicated to the secondary region.
- **NorthWatch**: monitoring must continue to work. It must be possible to see the health of Hyderabad services if the primary region's monitoring copy is lost.
- **KMS keys**: the DR region needs access to the keys required to decrypt backups. Signing keys held in an HSM must have a documented recovery process.
- **Communication tools**: the incident bridge, the emergency notification tool and Northstar Status must not depend only on the Mumbai region.

10. Worked Examples

Example 1: failed restore test. In the January quarterly test, the DBA restores the Nexus Platform tenant metadata database. The restore itself takes 1 hour 20 minutes, longer than the Tier-1 RTO of 1 hour. The test is marked failed. The service owner raises a NorthDesk action to use a larger restore instance and parallel log replay, due within 30 days. The re-test in February takes 38 minutes and passes.

Example 2: regional outage. At 14:05 IST on a Wednesday, the cloud provider reports a storage fault in Mumbai with no repair estimate. Tier-1 services are failing. The on-call engineer opens SEV1 at 14:08 and the Incident Commander becomes DR Commander. Because no repair estimate is available and a failover takes time, the DR Commander declares a disaster at 14:15. Northstar Status is updated by 14:35, within 30 minutes of the declaration. The Hyderabad replicas show a last applied log at 14:03, so the data gap is 2 minutes, inside the 15-minute RPO. Traffic is switched at 14:52 and normal metrics are confirmed at 15:05, so the achieved RTO is 50 minutes, inside the 1-hour objective.

Example 3: accidental table deletion. A data fix script deletes 12,000 rows from a Tier-1 reporting table at 11:20 IST. Failover would copy the damage, so the DBA performs a point-in-time recovery to 11:18, exports the 12,000 rows and applies them back after two-person review. No traffic is moved. The post-incident review follows ENG-004.

Example 4: old data request. A customer asks for data as it stood on a date 5 months ago. Daily backups are held for 35 days only, so only the monthly backup of that month is available, meaning the data is as at the first backup of that month. The service owner explains that restoring anything finer is not possible and records the request in NorthDesk.

11. Exceptions and Escalation

Exceptions to the recovery objectives, backup frequency or retention must be approved by the Head of SRE and recorded in NorthDesk with an expiry of no more than 90 days. Longer exceptions require the VP Engineering. Questions during a live disaster go to the DR Commander; outside an incident, questions go to SRE through NorthDesk.

If a disaster may involve a security breach, the DR Commander must inform the Security Desk (ext. 5555 or security@northstar-systems.example) immediately, and recovery steps must preserve forensic evidence, which is kept for 12 months. If the Bengaluru headquarters is also unavailable, the Crisis Management Team led by the COO coordinates, Bengaluru staff work remotely within 4 hours and critical teams operate from Mumbai. Regulatory time limits for breach notification are illustrative and to be verified by Legal before external use.

12. Frequently Asked Questions

What are the Tier-1 recovery objectives? RPO 15 minutes and RTO 1 hour.

How long do we keep daily and monthly backups? Daily backups for 35 days and monthly backups for 12 months.

When are DR drills held? Twice a year, in April and October.

How often are restores tested? Quarterly for Tier-1 databases.

Where are backups copied? To the secondary region in Hyderabad, encrypted.

13. Related Documents

- SEC-013 sets out the business continuity plan and crisis management structure that this runbook supports.
- ENG-009 covers database change management, including the rules for data fixes used in point-in-time recovery.
- SEC-005 covers encryption and key management used to protect backups.
- ENG-010 describes the monitoring that proves backups and recovery are healthy.

14. Revision History

Version	Date	Change
1.0	2024-10-15	First approved runbook
1.2	2025-02-24	Added drill reporting and dependency checks
1.3	2025-06-12	Clarified retention, copy to Hyderabad and failback steps